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Selection Committee
University of Copenhagen
Department of Computer Science (DIKU)
CopeNLU Lab
May 10, 2026

Subject: Application for PhD Position - Mechanistic Interpretability & LLM Security

Dear Members of the Selection Committee,

I am writing to formally submit my application for the PhD position within the CopeNLU group at the University of Copenhagen. As an AI Engineer specializing in advanced sequential architectures and trustworthy machine learning, I am deeply motivated by your project: "A Mechanistic Framework for Mitigating the Susceptibility of LLMs to Learning False Information."

Currently, I am completing my state engineering thesis at **Sonatrach**, where I am developing a **Generative Digital Twin** for critical industrial systems. My research addresses the "reality gap" by embedding physical constraints into neural networks using **Physics-Informed Neural Networks (PINNs)**. This experience has made me acutely aware of the dangers of models "learning" and propagating incorrect information—whether it is physically inconsistent sensor data or, in the case of LLMs, false factual information.

I believe my technical background in "Beyond-Transformer" architectures, specifically **Mamba-SSM** and **xLSTM**, provides a unique perspective for **Mechanistic Interpretability**. While most current interpretability research focuses on Attention mechanisms, I am interested in how state-space models compress and retrieve information. Understanding the "mechanics" of these linear recurrence systems could reveal new pathways for mitigating the absorption of false information that are not visible in standard Transformers.

The University of Copenhagen's reputation for excellence in NLP and the CopeNLU lab's focus on foundational challenges in trustworthy AI make it my first choice for doctoral research. I am a highly motivated, mathematically rigorous researcher, and I am eager to contribute to the security and transparency of next-generation language models.

I have attached my academic transcripts, my CV, and a research statement for your review. I would be honored to discuss my potential contributions to your group during an interview.

Thank you for your time and consideration.

Sincerely,

Ismail AISSAOUI